

Supplementation of a western diet with golden kiwifruits (*Actinidia chinensis* var.'Hort 16A':) effects on biomarkers of oxidation damage and antioxidant protection



Background The health positive effects of diets high in fruits and vegetables are generally not replicated in supplementation trials with isolated antioxidants and vitamins, and as a consequence the emphasis of chronic disease prevention has shifted to whole foods and whole food products. **Methods** We carried out a human intervention trial with the golden kiwifruit, *Actinidia chinensis*, measuring markers of antioxidant status, DNA stability, plasma lipids, and platelet aggregation. Our hypothesis was that supplementation of a normal diet with kiwifruits would have an effect on biomarkers of oxidative status. Healthy volunteers supplemented a normal diet with either one or two golden kiwifruits per day in a cross-over study lasting 2×4 weeks. Plasma levels of vitamin C, and carotenoids, and the ferric reducing activity of plasma (FRAP) were measured. Malondialdehyde was assessed as a biomarker of lipid oxidation. Effects on DNA damage in circulating lymphocytes were estimated using the comet assay with enzyme modification to measure specific lesions; another modification allowed estimation of DNA repair. **Results** Plasma vitamin C increased after supplementation as did resistance towards H₂O₂-induced DNA damage. Purine oxidation in lymphocyte DNA decreased significantly after one kiwifruit per day, pyrimidine oxidation decreased after two fruits per day. Neither DNA base excision nor nucleotide excision repair was influenced by kiwifruit consumption. Malondialdehyde was not affected, but plasma triglycerides decreased. Whole blood platelet aggregation was decreased by kiwifruit supplementation. **Conclusion** Golden kiwifruit consumption strengthens resistance towards endogenous oxidative damage.